

A Cutoff Criterion for Cancellation on the Plane

Abstract

We record a cancellation criterion for an integrable function on the plane. A uniform bound on the testing functions and an integrable flux-tail estimate imply that the source has zero mean. The criterion isolates a limiting argument relevant to weighted elliptic estimates; verifying its hypotheses for a compressible flow remains a separate problem.

1 The cancellation criterion

All integrals are taken with respect to Lebesgue measure on $\mathbb{R}^2$, and $|x|$ denotes the Euclidean norm. The cancellation argument in Fan–Lü–Wang [1, Lemma 3.3] motivates the following elementary analytic criterion.

Lemma 1.1 *Let $f, h \in L^1(\mathbb{R}^2)$ be real-valued, with $h \geq 0$ almost everywhere. Let $\chi_n : \mathbb{R}^2 \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be measurable. Suppose that constants $B, C \geq 0$ satisfy*

$$|\chi_n(x)| \leq B \quad (n \in \mathbb{N}, x \in \mathbb{R}^2), \quad \chi_n(x) \longrightarrow 1 \quad \text{for a.e. } x. \quad (1.1)$$

If, for every $n \in \mathbb{N}$,

$$\left| \int_{\mathbb{R}^2} f \chi_n \, dx \right| \leq C \left(\int_{\{|x| \geq n\}} h \, dx \right)^{1/2}, \quad (1.2)$$

then $\int_{\mathbb{R}^2} f \, dx = 0$.

Proof Put $E_n = \{x \in \mathbb{R}^2 : |x| \geq n\}$ and $H_n = \int_{E_n} h \, dx$. Each E_n is measurable. Since $h \in L^1(\mathbb{R}^2)$, its restriction to E_n is integrable, and $H_n \geq 0$. For each fixed x , the indicator $\mathbf{1}_{E_n}(x)$ is eventually zero. Hence

$$h \mathbf{1}_{E_n} \longrightarrow 0, \quad |h \mathbf{1}_{E_n}| \leq |h| \in L^1(\mathbb{R}^2).$$

The dominated convergence theorem gives

$$H_n = \int_{\mathbb{R}^2} h \mathbf{1}_{E_n} \, dx \longrightarrow 0. \quad (1.3)$$

On the other hand, each $f \chi_n$ is measurable up to a null set and

$$|f \chi_n| \leq B |f| \in L^1(\mathbb{R}^2).$$

Thus all testing integrals in (1.2) are integrals of integrable functions. By (1.1), $f \chi_n \rightarrow f$ almost everywhere. A second application of dominated convergence yields

$$I_n := \int_{\mathbb{R}^2} f \chi_n \, dx \longrightarrow I := \int_{\mathbb{R}^2} f \, dx. \quad (1.4)$$

Combining (1.2) and (1.3), we obtain

$$0 \leq |I_n| \leq C \sqrt{H_n} \longrightarrow 0.$$

Equation (1.4) also implies $|I_n| \rightarrow |I|$. Uniqueness of the limit gives $|I| = 0$, as asserted. $\square$

Remark 1.1 The estimate (1.2) is a substantive hypothesis. An application to a momentum equation must establish it from admissible testing, integrability, and control of the flux at infinity. Lemma 1.1 does not assert that these requirements hold for a particular compressible solution, nor does it establish a global existence theorem.

References

1. X. Fan, B. Lü, X. Wang, Hardy Space and Exponential Decay of Inhomogeneous Navier–Stokes System in $\mathbb{R}^2$, arXiv:2610.04884v1 (2026).